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Food system planning within Rama First Nation: a strengths-based participatory approach to promote food sovereignty

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Introduction: Through community-based initiatives, Indigenous Communities are leading efforts to revitalize their food systems for food security and food sovereignty. In support of these ongoing Community actions, we worked with the Chippewas of Rama First Nation, one of seven Williams Treaties First Nations in Ontario, Canada, to develop a Community-informed inventory of existing food assets to inform food system planning. Through engagement with Community members and leaders, we aimed to (1) identify existing food-health related programs and resources; and (2) describe Community perspectives on opportunities to improve access to healthy, local, and traditional foods as pathways to food security and food sovereignty.

Methods: Guided by Two-Eyed Seeing and community-based participatory research principles, a mixed methods approach incorporating a strengths-based survey and key informant conversational interviews with Rama First Nation Community members was undertaken. Descriptive statistics were used to summarize the survey data and generate frequency distributions of participant responses. An inductive thematic analysis was applied to identify codes and themes from the open-ended questionnaire responses and key informant interviews.

Results: A total of 99 Community members participated in the survey. Food programs and land-based practices, along with barriers that impede access, availability, utilization, and stability of the Community food system, informed emergent themes. Eleven themes were identified across four key areas: (1) identifying opportunities to address barriers to food security, (2) reconnecting with local and traditional foods for health and wellbeing, (3) expanding on existing Community-led pathways to food access and cultural connection, and (4) mobilizing Community assets to strengthen the local food system.

Discussion: Land-based and traditional food acquisition practices (hunting, fishing, foraging, and preparation) may promote access to healthy foods while fostering cultural identity and well-being. Creating Community-led opportunities to learn and practice traditional skills are important for retaining cultural knowledge and rebuilding connections to traditional food systems. The knowledge generated can inform ongoing planning and shared decision-making to strengthen the food systems of Williams Treaties First Nations.

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KEYWORDS

community-based participatory research, food security, food sovereignty, food systems, indigenous peoples, strengths-based

1 Introduction

Indigenous peoples have lived on Turtle Island (North America) for thousands of years, and their food systems are deeply connected to the land and waters (Atleo and Boron, 2022; Lemke and Delormier, 2017). For generations, the food systems of many First Nations across Canada were composed of traditional foods hunted or collected from the natural environment, supporting both health and cultural identity (Batal et al., 2021b). These practices remain intrinsic to First Nations culture and are woven into all aspects of life (Batal et al., 2021b). However, colonial-influences have disrupted Indigenous food systems, reducing the availability, quality, safety, and access to traditional foods (Joseph and Turner, 2020; Kuhnlein and Receveur, 1996; Stanley et al., 2020). Factors such as land privatization and government regulations have led to forced relocation from their homelands, environmental degradation, and loss of land and water resources (Malli et al., 2023; Mead et al., 2010; Richmond and Ross, 2009). The impacts include the loss of traditional knowledge for hunting, gathering, or preparing food from the land, and a growing disconnect from traditional foods (Kuhnlein and Receveur, 1996; Settee and Shukla, 2020). Furthermore, the introduction of Westernized foods and dietary transition has contributed to poor nutritional quality (Shukla et al., 2019; Willows, 2005; Kuhnlein et al., 2004; Shafiee et al., 2024). These barriers, accompanied by high food prices (Shafiee et al., 2024; Turner et al., 2013), low income (Batal et al., 2021a,b; Tarasuk et al., 2021), lack of on-reserve grocery stores (Richmond et al., 2021), lack of transportation (Batal et al., 2021b; Socha et al., 2011), and exposure to environmental contaminants (Turner et al., 2013; Juric et al., 2018; Seabert et al., 2014), have contributed to food insecurity.

In Ontario, Canada, the prevalence of food insecurity[1] has been disproportionately felt by Indigenous households (Batal et al., 2021a; Willows et al., 2024; Shafiee et al., 2022). On-reserve First Nations households have been found to experience food insecurity at elevated levels (ranging from 24 to 60%) that are 3 to 5 times greater than non-Indigenous Canadian households (12%) (Batal et al., 2021a; Chan et al., 2021). This health inequity has been linked to diet-related health conditions such as obesity, type 2 diabetes, cardiovascular disease, and cancer (Kuhnlein et al., 2004; Domingo et al., 2021b; Haman et al., 2010; Elias et al., 2019; Egeland et al., 2011; Bruce et al., 2014). Many initiatives have emerged to address health challenges associated with food insecurity while promoting community-led and self-determined food systems. Community gardens, school food programs, greenhouses, and food preparation and processing workshops are frequently cited examples of food- and health-promoting community-based initiatives (Bagelman et al., 2016; Tsuji et al., 2020; Wesche et al., 2016; Lamalice et al., 2018; Thompson et al., 2011). These efforts are also often part of larger food sovereignty endeavours, which focus on enhancing community autonomy to define food systems in support of well-being and self-determination (Coté, 2016; Cidro et al., 2015; Kepkiewicz and Dale, 2019). While collaboration with community leaders is highlighted as a key feature of these programs, the process of facilitating inclusive and participatory engagement is not always well described. A better understanding of approaches grounded in receiving community input and guided by community preferences could improve community-partnered research to strengthen community food systems (Trott and Mulrennan, 2024; Doustmohammadian et al., 2022). This paper describes the participatory research process undertaken and the findings from a Community-led strengths-based assessment to support food system planning as part of an initiative in the Chippewas of Rama First Nation, a Community[2] located in central Ontario.

Rama First Nation is one of the seven Williams Treaties First Nations that has identified access to year-round healthy and affordable food as a priority. The seven Nations comprise a collective of Anishinaabe Communities, which includes Ojibwe and Mississauga Communities, across southern and central Ontario. These include Ojibwe Communities of Rama, Beausoleil, and Georgina Island First Nations, and Mississauga Communities of Alderville, Curve Lake, Hiawatha, and Scugog Island First Nations. Together, these Nations share a treaty relationship shaped by the 1923 Williams Treaties and the 2018 settlement agreement that reaffirmed harvesting rights and responsibilities (Murray, 2023; Surtees, 1986). Each Community maintains its own governance but collaborates on shared priorities. In recent years, several Williams Treaties First Nations have advanced Community-led food initiatives, with Rama First Nation, in particular, leading opportunities for knowledge exchange and resource sharing across the Communities in support of collective food sovereignty (Domingo et al., 2021a).

While Rama's 10-year Community Visioning Process (About Rama, 2020) identifies food and health-promoting activities as a strategic priority, it has yet to develop a comprehensive, Community-informed inventory of existing food assets and priorities to strengthen the local food system. This could inform decisions that promote engagement with healthy foods and the overall wellness of the Community. Our project, therefore, builds on this work through a community-led approach to support ongoing efforts to strengthen the local food system for food security and food sovereignty. We engaged with Community members, leaders, and Elders within Rama First Nation to (1) identify existing food-health related programs and resources; and (2) describe Community perspectives on opportunities to improve access to healthy, local, and traditional foods as pathways to food security and food sovereignty. By documenting existing food assets, programs, and skills, we highlight the participatory process

1 According to the Food and Agriculture Organization (FAO), food security exists when people, at all times, have physical and economic access to sufficient, safe, and nutritious food to meet their dietary needs and food preferences for an active and healthy life (Food and Agriculture Organization of the United Nations, 2008).

2 Based on guidance from Community advisors (K-A.C. and T.H.), 'Community' is used throughout to refer to Indigenous Communities and differentiate from non-Indigenous communities.

taken and the strengths within the Community that can be uplifted to improve access to healthy foods. Our Community-led, strengths-based approach contributes to the literature on participatory food system transformation and advances calls to support Community self-determined food systems and culturally relevant responses for access to nutritious and traditional foods for health (i.e., United Nations Sustainable Development Goals 2 and 3; United Nations Declaration on the Rights of Indigenous Peoples Articles 23 and 24) (United Nations Department of Social Economic Affairs, 2025; UN General Assembly, 2007). In addition to supporting Rama First Nation, the findings may also assist other Communities within the Williams Treaties Territory with food system planning, including opportunities for resource-sharing and collective action towards food sovereignty.

2 Materials and methods

2.1 Research approach

Drawing on Indigenous methodologies of relationality (Kovach, 2021) and community-based participatory research (CBPR) (Drawson et al., 2017), we collaborated with Rama First Nation to support ongoing efforts to strengthen the food system. This community-based participatory project builds on engagements initiated in 2016 as part of the Crown-Indigenous Relations and Northern Affairs Canada (CIRNAC) funded initiative to support Climate Change Adaptation planning with Indigenous communities (Warren et al., 2020). The work is now part of a larger initiative, Minwanjigewin (Anishinaabemowin word for "to eat something that tastes good and is good for you"), which aims to identify and build on Community-assessed strengths to inform the development of an Indigenous-led, Community-based food hub with the Williams Treaties First Nations. Guided by Etuaptmumk [Two-Eyed Seeing (Martin, 2012)], the project team consisted of Indigenous (T.H., K-A.C., M.W.), racialized (A.D.), and non-Indigenous (J.A., K.S.) members working together in mutual ethical space of respect and reciprocity (Restoule, 2008).

Two-Eyed Seeing, as described by Mi'kmaw Elder Albert Marshall, encourages the bridging of knowledge between traditional Indigenous knowledge and Western science while recognizing that one perspective may further specific concepts and insights better than another (Martin, 2012; Datta, 2017; Wright et al., 2019). This guiding principle asserts that both Indigenous and Western ways of knowing are essential for meaningful collaboration (Bartlett et al., 2012). Our collaboration was further guided by the First Nations principles of Ownership, Control, Access and Possession (OCAP®)[3] to support Community governance of information (Robson et al., 2017). There were various ways the OCAP principles were integrated, such as sharing learnings with the Community and supporting opportunities for Community members to provide feedback to guide project processes and outcomes. Guidance from Community leadership and the Community Elders ensured these principles were applied by providing direction on appropriate engagement, sense-making, and knowledge translation during all phases of the research. To further aid in our ethical engagements, the Seven Generations Principle (Clarkson et al., 1992), a Haudenosaunee teaching that has been adopted by many Indigenous Communities, provided a guide to consider the project's larger scope of learning from embodied experiences and history, while considering the impact on future generations.

CBPR principles (Drawson et al., 2017; Israel et al., 2010) were also applied to facilitate a process in which Community advisors (K-A.C., T.H.) guided the project's direction. This consisted of taking a collaborative approach to prioritize Community needs, while also fostering reciprocal learning in Indigenous knowledges and science, capacity building in research, shared decision-making (co-creation of the research scope), and Community ownership of the project. As advised by the Community leader (K-A.C.), two Community gatherings were held to share findings from this project, with future gatherings in the planning stages to foster relationship building. In alignment with these principles, we approach relational accountability as a practice of acknowledging Truth and Reconciliation enacted through centering Community voices, and sustaining respectful, reciprocal partnerships. All co-authors participated in project planning, analysis, writing, and editing. T.H. (primary co-author), a member of Rama First Nation, led knowledge gathering and collaborated with J.A., (primary co-author) and M.W., on analysis and synthesis of findings as a graduate student. The team brought diverse perspectives and expertise in health sciences, participatory approaches, and food systems: J.A. is a graduate student with knowledge on nutritional-health related research, M.W. is a biomedical science student and member of Wikwemikong First Nation, K.-A.C. is a member of Georgina Island First Nation and holds extensive knowledge in climate change adaptation and food sovereignty, K.S. and A.D. are applied public health researchers, with training in program evaluation, community-based research, and implementation science. Together, the project team remains committed to contributing to scholarship that centers Indigenous knowledge systems, promotes food sovereignty, and supports the health and well-being of Indigenous Communities. All activities were conducted in accordance with the Tri-Council Policy Statement (TCPS2) on Ethical Research Involving First Nations, Inuit, and Métis Peoples of Canada (Government of Canada IAP on RE, 2019) and were reviewed by the University of Guelph Research Ethics Board (REB #1203).

2.2 Community context

Rama First Nation is located in central Ontario, approximately 10 kilometers east of Orillia. The Community spans approximately 2,350 acres, situated along the shores of Lake Couchiching and Black River (Figure 1). In 2021, Rama had a registered population of about 2,100 members, with roughly 800 living on reserve in approximately 450 households (Government of Canada, Indigenous Services Canada, 2022). Rama is accessible year-round by a county road which runs through the Community leading into the central area. The nearest full-service grocery store is located in the City of Orillia. However, within the Community itself, food access remains limited as year-round access to fresh and healthy foods typically requires members to travel off reserve.

Rama First Nation has been historically recognized as "the gathering place," where travelers rested before continuing their journey, where meetings were held, and agreements were signed (Chippewas of Rama

3 OCAP® is a registered trademark of the First Nations Information Governance Centre (FNIGC). For a full definition please refer to https://fnigc.ca/ocap-training/.



FIGURE 1
Map of communities of the Williams Treaties First Nations. The map was created by collaborators at the Toronto climate observatory on May 6, 2026.

First Nation, 2020). The Chippewas of Rama First Nation are known as Keepers of the Fish Fence, caretakers of the Mnjikaning Fish Weirs, an ancient network of wooden stakes used to sustainably trap and harvest fish in the narrows between Lake Couchiching and Lake Simcoe (Ringer, 2006). These weirs are estimated to be over 5,000 years old and represent a body of ancestral knowledge related to engineering, seasonal harvesting, and ecological balance (Mulligan, 2003). Although traditional food acquisition practices have declined due to urbanization and land use changes, they remain important as many Community members continue to engage in land-based activities. In recognition of active leadership in food sovereignty initiatives in Rama First Nation and interests in improving access to year-round, healthy foods on reserve, this paper showcases the strengths-based approach to support the Community in strengthening the local food system. The Community is actively pursuing strategies to improve food security, chronic disease prevention, and Community wellness through land-based education, local food programs, and traditional knowledge revitalization.

2.3 Research participation

2.3.1 Participant recruitment

All members of Rama First Nation, 18 years of age and older, were eligible to participate in the project. Participants were recruited through purposive and snowball sampling techniques to inform diverse insights relevant to the project objectives and enhance participation through social connections (Tajik et al., 2024; Ting et al., 2025; Naderifar et al., 2017). Purposive sampling is widely recognized as an appropriate strategy for selecting Community members who can generate information-rich experiences and perspectives on the priorities and needs of the Community (Tajik et al., 2024; Palinkas et al., 2015). For the purposes of this project, Rama First Nation defines a Community member as a registered member, the spouse or child of a registered member, or a staff member of the organization. Given the project's community-based nature and the information being sought, all individuals who met this definition were encouraged to participate without additional restrictions. Knowledge was gathered from both on and off-reserve members to ensure that the perspectives of all Community members are represented. Additional insights were informed by members who work in areas connected to food such as staff, Community leaders, and decision makers who can provide an understanding of how current food initiatives operate and where barriers may exist.

A strengths-based survey and conversational key-informant interviews (Kovach, 2010) were used to provide a comprehensive view of Community-identified priorities and inventory of food programs and services. Community input from a food strengths survey with key informant conversational interviews was intended to provide a comprehensive understanding of healthy food access and engagement.

2.3.2 Strengths-based survey

The questionnaire was titled *Minwanjigewin Survey - Nourishing our Communities through Sustainable Food Systems*

(Supplementary File 1). The survey was adapted from a pre-existing instrument developed and administered in the Chippewas of Georgina Island First Nation, one of the Williams Treaties First Nations Communities. The tool was revised to reflect the specific context of the Chippewas of Rama First Nation Community. The questionnaire followed a strengths-based approach (Prehn, 2025) with the intent to highlight existing Community assets, traditional food-related knowledge and practices, and priorities related to the Community food system and food access. The survey comprised both open- and closed-ended questions to allow for the collection of quantifiable information while also capturing personal reflections. Content was organized by categories, including (1) participant characteristics, (2) the current Community food system, (3) envisioned changes to the food system, (4) existing Community assets and resources, and (5) Community-informed actions. Topics explored within the questionnaire included (1) perceptions of food system challenges and requested changes to the current food system; (2) traditional diets, including interest in returning to traditional ways of eating, frequency of consumption, Community-defined components of such diets, and barriers to access; and (3) engagement in traditional skills and practices, including participation in, access to resources, barriers to engagement, and interest in supports or programming to enhance involvement.

The questionnaire was administered in English between February 1, 2025, and March 14, 2025, using both online and paper-based formats to increase accessibility and participation. The online version was shared through Community social media channels, and paper copies were distributed through key Community locations and events. The introduction of the survey outlined the purpose of the study, and informed consent was obtained prior to participation. Participants were informed that their participation was voluntary, and the questionnaire was expected to take between 20 to 45 min to complete, depending on the level of detail provided.

2.3.3 Conversational interviews

To complement the questionnaire responses and gain deeper insights into Community perspectives, semi-structured interviews (Kilian et al., 2019) were conducted with key informants identified through purposive sampling (Tajik et al., 2024; Palinkas et al., 2015). Key informants (n = 4) included Rama First Nation staff and Community members with lived experience or specialized knowledge in areas such as the local food system, traditional practices, nutritional-health services, and Community development. This approach, combined with the survey, generated information-rich and diverse perspectives relevant to the study objectives to achieve information power (Malterud et al., 2016).

Semi-structured conversational interviews drew on survey questions and were tailored to each participant's roles, experiences, and areas of expertise. Our approach aligned with Indigenous methodologies to uphold the relational dynamics of Indigenous ways of knowing through storytelling (Kovach, 2010; Kilian et al., 2019; Wilson, 2001). Key topics included roles within the community, existing programs, services and resources for food access; approaches to knowledge sharing and collaboration; and perspectives on goals and strategies for improving healthy food access and food sovereignty. All interviews were conducted in person by a Community student researcher (T.H.) familiar with the Community context. Interview notes were taken with participant consent. Prior to each interview, participants were provided with a plain-language overview of the study, and written informed consent was obtained. The knowledge gathered was anonymized during transcription and stored securely by Cambium Indigenous Professional Services (T.H.) to maintain confidentiality.

2.4 Research analysis

2.4.1 Qualitative analysis

An inductive thematic analysis approach was applied to identify codes and themes within the qualitative data derived from both the open-ended questionnaire responses and key informant interviews (Braun and Clarke, 2021). The analysis followed a multi-stage, inductive, and data-driven approach guided by the Indigenous methodologies employed by Margaret Kovach, which outlines 4 steps: *First-Level Analysis, Relational Analysis, Intentional Interpretation and Theme Development* (Kovach, 2021). This approach acknowledges the relational, circular, and reflective nature of Indigenous knowledge systems and centers participant voices throughout the process.

The process began with members of the team (T.H. and J.A.) identifying "touchstones" to ensure the analysis remained rooted in the intent and context of the research, with emphasis on lived experience, community self-determination, and relational accountability (Kovach, 2021). In the *First-Level Analysis*, interview and open-text survey responses were independently reviewed and openly coded by two analysts, generating initial codes that captured meaningful concepts. The team then met to compare and discuss their individual coding frameworks, developing a shared structure. This was followed by *Relational Analysis*, in which axial coding was used to explore relationships between codes, allowing the grouping of related concepts into broader thematic categories. The analysis reflected an iterative process whereby the coding framework evolved as data were re-coded and reorganized to clarify categories and ensure responsiveness to emerging insights. In the stage of *Intentional Interpretation*, the research team engaged in reflective dialogue to critically assess whether the themes under development authentically represented the voices and experiences expressed by the Community respondents. Finally, during *Theme Development*, overarching themes were constructed from refined categories, with thematic labels reflecting interpretive summaries grounded in participant narratives (Kovach, 2021). Nvivo software version 14 Pro (QSR International) was used for coding and analysis.

2.4.2 Quantitative analysis

Quantitative data from the close-ended survey questions were imported into Microsoft Excel for analysis. Descriptive statistical analysis was conducted to summarize the data and generate frequency distributions of participant responses. Frequency counts and percentages were calculated to summarize the data, using charts and graphics to visualize the findings for interpretation and comparison of response patterns. These frequencies were used to provide an overview of responses related to food access, food programs, and

Community-identified priorities. The graphical summaries will support the identification and inventory of existing food programs and services, as well as the assessment of Community priorities and initiatives related to food security and sustainability.

3 Results

Of approximately 2000 total Rama First Nation Community members, 99 members participated in the survey, and their ages ranged from 18 to 74 years. The interview and survey participants were all Community members of Rama First Nation with different living experiences, either residing within or outside the Community. Participants varied in their sex, age, and living situations as outlined in Table 1.

The integrated analysis of the qualitative conversational interviews and open-ended survey questions, combined with the closed-ended quantitative survey responses, informed four key areas: (1) identifying opportunities to address barriers to food security, (2) reconnecting with local and traditional foods for health and wellbeing, (3) expanding on existing Community-led pathways to food access and cultural connection, and (4) mobilizing existing Community assets to strengthen the local food system. Themes (n = 11) provide insight into existing food-health related programs, resources, and opportunities to improve food access and are listed in Table 2.

TABLE 1 Survey participant demographics.

Demographics	Number of respondents	Percentages (%)
Total participants	99	
Sex (n = 96)		
Female	76	79
Male	20	21
Age distribution (n = 94)		
18–24	6	6
25–34	25	26
35–44	19	20
45–54	18	19
55–64	21	22
65+	5	5
Living situation (n = 92)		
I live off reserve	29	32
I live on reserve	63	68
Living experiences (n = 95)		
I have lived in other Indigenous communities	9	9
I have lived in rural communities	35	37
I have always lived in my community	36	38
I have lived in urban communities	50	53

3.1 Identifying opportunities to address barriers to food security

Many participants (91%) reported that produce was highly accessible, whereas local and traditional food sources of meat, foraged foods, and fish were less accessible (Figure 2). This was consistent with participants reporting high engagement in gardening as a traditional practice (48%) while other practices, including hunting/trapping (6%), gathering/foraging (38%), and fishing/harvesting crustaceans (22%), had comparatively lower engagement. While participants expressed strong interest in accessing locally sourced foods, when asked about the possibility of transitioning to a more traditional diet, participants expressed uncertainty. The main barriers to accessing traditional food that were identified by participants included the loss of traditional knowledge and skills (60%), and concerns about the high cost of food (42%). These concerns highlighted several challenges across the four pillars of food security: availability, access, utilization, and stability of their local food system (Shafiee et al., 2022).

3.1.1 Variety of fresh produce from local sources

A common concern raised by participants was the limited variety of fresh produce from their local supplies. The availability pillar of food security considers the supply of food. This includes foods produced or brought into the Community, as well as the presence of safe and healthy food (Shafiee et al., 2022). The Community's main supply of fresh produce on reserve is derived from the Community farm, and a local market which resides in the Community's convenience store.

TABLE 2 Themes providing insight into existing food-health related programs, resources, and opportunities to strengthen the local food system.

Key areas	Themes
Identifying opportunities to address barriers to food security	• Variety of fresh produce from local sources • Affordability of local foods impeding access • Loss of traditional food knowledge and skills as a barrier to utilization • Concerns for the long-term sustainability of the local food system
Reconnecting with local and traditional foods for health and wellbeing	• Restoring traditional food knowledge and skills • Strengthening infrastructure and resource access for traditional food practices
Expanding on existing Community-led pathways to food access and cultural connection	• Promoting nutrition education and healthy eating • Addressing financial barriers to healthy food access • Promoting holistic wellness through land-based practices
Mobilizing existing Community assets to strengthen the local food system	• Revitalizing land and natural food sources within Community • Fostering shared stewardship with Williams Treaties First Nations for sustainable food systems

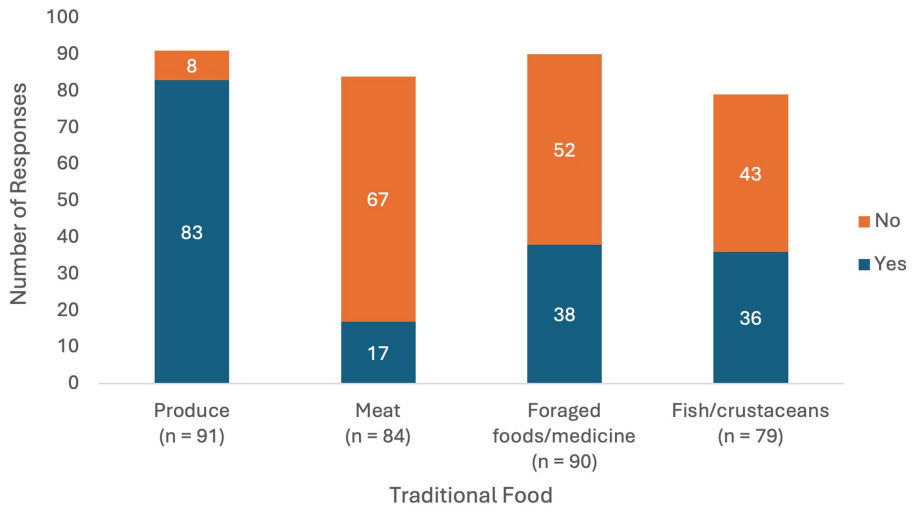


FIGURE 2
Current access to local and traditional food sources.

When participants were asked about their local food access from each of these sources, many highlighted that although fresh produce is highly accessible, there are ongoing challenges that limit both quantity and quality of food: *"The quality and diversity of the food from the Community garden* [Community farm] *is lacking"* (survey respondent 1). Participants linked their concerns with the availability and variety of fresh produce to factors such as having limited control over their land and resources due to government restrictions: "*the government and its restrictions and loss of traditional harvesting areas is our largest barrier"* (survey respondent 2). Many highlighted that this has disrupted the harvest of fresh produce, including berries, vegetables, and medicinal herbs. Additionally, participants noted that seasonal conditions can limit what is grown at the farm, impacting year-round availability. These factors can serve as barriers to the availability, variety, quality, and safety of fresh produce from local sources.

Participant concerns over food availability also extended to store-bought foods. While the Community's farm is an important source of fresh produce and a grocery store for the Community is in the planning stages, a small convenience store remains a significant source of food on reserve, offering select products, from fresh produce to canned goods. However, Community members have found the options for fresh produce to be limited from this source as well: *"Rama Farm provides local produce at the Country Market,* [Shell Gas Station] *but it is hit or miss. Some days there is lots and other days everything is sold out"* (survey respondent 6). Participants attributed this concern to the lack of a local grocery store for access to both local and traditional foods. The absence of such infrastructure was perceived as a major barrier to obtaining local produce, contributing to limited food choices in the Community: *"A proper grocery store for all of our locally source foods would be great. And we could add other items we cannot source"* (survey respondent 4). These insights further underscore the Community's need for improved, sustainable, and locally available food infrastructure.

3.1.2 Affordability of local foods impeding access

Consistent access to traditional/country foods through traditional harvesting methods, alongside access to market foods have been identified as important facilitators of local food access (Power, 2008). A key barrier to reliable access to local and traditional foods, as emphasized by participants, was affordability. As described previously, the main sources of local produce are the Rama Community farm and the local market at the convenience store gas station. While these sources have been identified as helpful to the Community, challenges with high prices of fresh produce and other store-bought foods are compromising the Community's food access. Many highlighted that if the items sold locally were more affordable, then they would be able to purchase fresh produce more often. One participant explained, *"the vegetable stand and food items at the Shell* [local market] *are very expensive. I would shop more here if it was more affordable"* (survey respondent 13). The Community enjoys supporting their local food vendors, such as the Community farm, but the expense often prevents them from doing so. During one of our conversational interviews, this issue became especially apparent when discussing economic disadvantages as a barrier for accessing high quality foods disrupting health outcomes throughout the Community: *"Food security is a big issue in Community. I just had someone the other day tell me that their doctor told them they needed to eat more fruits and vegetables. But then that person shared with me, 'I cannot do that, I cannot afford it'"* (key informant 1). This suggests a need for community-based programs that can provide a financial subsidy to improve access to healthy foods to support wellbeing.

3.1.3 Loss of traditional food knowledge and skills as a barrier to utilization

The utilization pillar of food security considers adequate dietary intake, as well as factors such as cultural foods, food safety, food storage, and food literacy (Shafiee et al., 2022). A lack of traditional food knowledge, food preparation skills, and nutrition can negatively impact opportunities to use traditional foods: *"A lot of the stuff that is grown at the farm, people do not know what to do with"* (key informant 1). Concerns about a lack of awareness of healthy foods and of what constitutes a traditional diet were highlighted as factors contributing to misconceptions about traditional foods, which may limit engagement with traditional foods. As this loss of knowledge has been recognized as a contributor to dietary changes, many emphasized the need for a shift towards traditional practices for better health: *"There*

needs to be a huge cultural shift […] *people in Community pride themselves on eating greasy foods that are claimed to be traditional. Scones. Fried bologna. All of the things we were raised on. It is difficult to find anything other than greasy food trucks at a powwow*" (survey respondent 3).

Participants expressed that their families once had deep knowledge about traditional foods, nutrition, and traditional practices. However, the passing of this knowledge over generations has been lost, impacting engagement in harvesting activities and disrupting intergenerational knowledge sharing. Participants highlighted the importance of fostering motivation and interest in preparing traditional foods: *"There is a lack of motivation and access to creative and efficient ways to source and prepare meals, but also, better knowledge of health benefits of traditional diet would promote motivation*" (survey respondent 5). Supporting opportunities for Community members to share their traditional knowledge with others was also discussed as a way to restore interest in a traditional diet.

3.1.4 Concerns for the long-term sustainability of the local food system

Food stability can be supported when everyone at the individual, household, and community levels have sustained access to safe, sufficient, and healthy foods at any given time (Shafiee et al., 2022). When Community members were describing the barriers to engaging with traditional practices, or consuming local and traditional foods, several expressed concerns about the stability of the Community's food system. It was shared that the traditional diet in general is a sustainable way of living, reflecting the long-standing Indigenous practice of using the resources available and ensuring nothing goes to waste: *"Our sustainable ways of living and cooking is a necessity to our practice*" (survey respondent 7). However, concerns about resource depletion and pollution are among a few factors that the Community has identified as threatening sustainable practices. One participant described their concerns about the lake water quality impacting the safety of the harvested fish: *"The waters around here are too polluted and too warm. Some fish have tumors, so I have a personal need to see them caught to know it's safe to eat*" (survey respondent 12). Participants also expressed concerns for wildlife and vegetation in the Community, as some feel the need to conserve the animals and other plant resources, preventing them from engaging in practices of hunting and harvesting: *"I feel like although our culture used to thrive on the land it seems in Rama if there are deer or moose one should not kill it because it is very rare to see […] here in this Community of Rama First Nation it feels the resources are scarce as they are*" (survey respondent 10). These insights highlight the impacts of environmental change on the traditional food system, as traditional food practices have become more difficult to maintain under changing local conditions.

3.2 Reconnecting with local and traditional foods for health and wellbeing

Revitalizing the food system to reconnect with local and traditional foods was identified as a key priority for promoting the health and wellbeing of the Community. Not only was access to healthy and affordable local produce prioritized (97%) but also reconnecting with traditional food practices by revitalizing food knowledge and preparation skills (74%) (Figure 3). When Community members were asked what supports they would need to engage with traditional foods, the most requested support was learning the skills and acquiring access to equipment for each traditional food practice (Figure 4). Each of these resources to acquire food from the land were emphasized as important to reclaim lost knowledge and provide opportunities for Community members to use traditional foods to support their wellbeing.

3.2.1 Restoring traditional food knowledge and skills

Educational programs to build traditional food knowledge and skills were a key resource voiced by the Community. Programs that aim to enhance skill development can enable Community members to relearn and practice their traditional ways of living. Participants described opportunities to embed nutrition education within skill-based programs that could further Community engagement and reconnection to traditional practices. One participant explained that *"food education* [within cooking workshops] *needs to be more than teaching the food groups, it must be teaching the importance of protein, and what carbohydrates do to the body*" (survey respondent 15). This not only reinforces the Community's desire to reconnect with traditional foods, but also to enhance their knowledge and understanding of a healthy diet. Many participants emphasized that food sharing activities such as Community freezers, gardens, and food-related

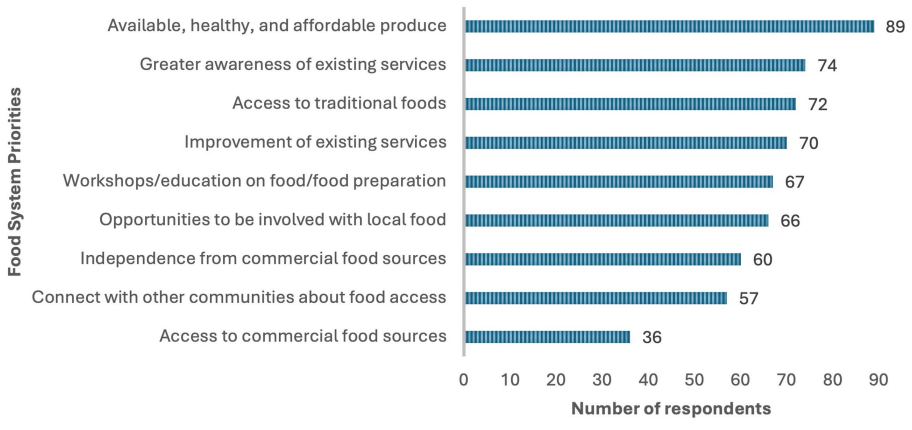


FIGURE 3
Community identified food system priorities (n = 91).

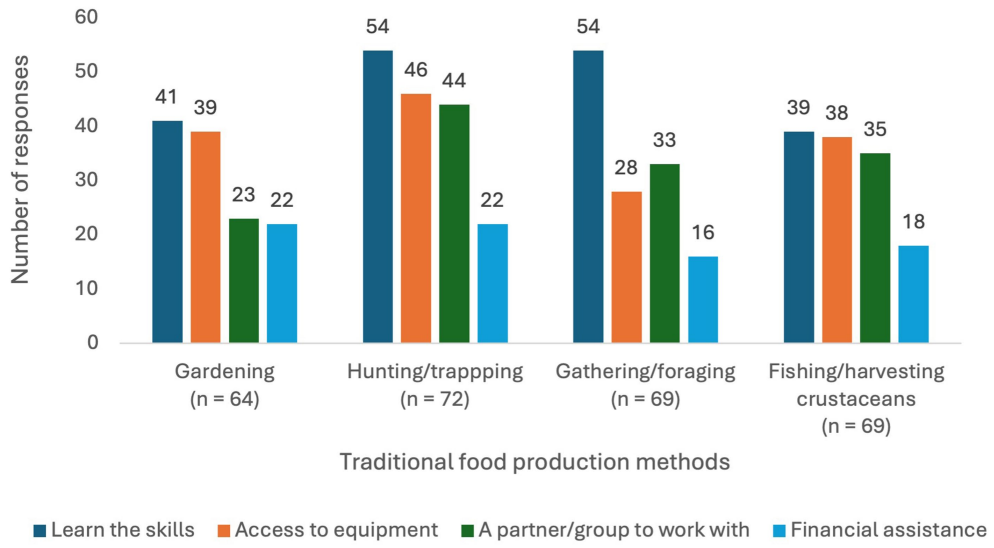


FIGURE 4
Key supports identified by community to engage with traditional food practices.

programs are opportunities to bring the Community together to re-instill and teach one another traditional food skills and knowledge: *"We should have group foraging, hunting, and fishing programs, for proper education on correctly gathering and foraging in and around Community"* (survey respondent 8). Furthermore, Community-oriented programs stress the importance of restoring knowledge sharing among the Community as an essential practice for cultural connection. During one of our conversational interviews, our key informant shared that this practice should be integrated within educational programming by involving Elder's within planning: *"It is essential to identify opportunities where Elders and members can share stories from growing up within all programming initiatives"* (key informant 4). Opportunities to learn from one another, particularly through the involvement of Elders and knowledge carriers, were identified as critical to maintaining practices passed down for generations that continue to shape the Community-guided food system.

3.2.2 Strengthening infrastructure and resource access for traditional food practices

Supporting traditional food knowledge and skills also requires physical resources such as appropriate equipment to engage in traditional practices, which was another key support requested by the Community (Figure 4). Participants shared that strengthening access to physical resources such as fishing and hunting equipment may help improve both access to and availability of traditional foods within the Community. In addition to harvesting tools, some Community members also identified resources for traditional food preservation practices: *"I would like to see an extra-large inbound root cellar, [and] a proper yet primitive meat locker"* (survey respondent 12). Traditional food preservation tools and practices highlight the Community's desire to enhance food availability and support year-round access to certain foods. This further illustrates the Community's interest in developing sustainable, locally driven approaches that support engagement with traditional foods and practices. Participants linked the importance of accessible equipment to fostering knowledge sharing within the Community towards improving the health and wellbeing for future generations: *"I would like to see a large food dehydrator at the farm so everyone in Community can be involved"* (survey respondent 9).

3.3 Expanding on existing community-led pathways to food access and cultural connection

Community members and key informants revealed existing programs and initiatives geared towards nourishing Community and strengthening connection to cultural practices for a sustainable food system. Table 3 outlines existing programs that promote healthy eating, improve food access, and support cultural and land-based learning within the Community.

3.3.1 Promoting nutrition education and healthy eating

While the Community identified a need for educational programs for building traditional food knowledge, several existing nutrition programs already promote engagement with healthy foods and knowledge sharing. Participants emphasized that current programs aim to promote exposure to healthy foods through Community initiatives: *"We host a Community Breakfast Club to offer healthier food options"* (key informant 1). Similar food engagement is offered at the Community elementary school by providing nutritional meal plans that incorporate traditional foods for students. A conversation with one of our key informants emphasized the importance of engaging youth in healthy eating in fostering lifelong habits: *"We know that when kids can be more involved in prepping something or cooking something, they are more apt to try it. Or if they see their peers trying it, [this] also allows them to try some different foods that they might not get at home, just to see if they like it"* (key informant 1). Programs in the Community not only facilitate learning about healthy eating but also introduce the Community to new types of foods and modes of food preparation. One

TABLE 3 Inventory of existing programs and initiatives.

Programs and initiatives	
Elementary school cooking classes	Freezer meal workshops
Secondary school cooking classes	Community breakfast club
Elementary school lunch & snack program	Compassionate glucose monitoring sensor programming
Youth hatchery program	Protein meal replacement distribution
Elder-led peace garden	Soup Swap
Community gardening	Food programming policies
Wild rice planting	Community freezer
Medicine walks	Medicine planting program
Pumpkin picking	Land-based learning survival course
Seeding workshop	Guided fishing trips
Good food box subscription	

participant shared: *"recreation hosts a soup swap,* [and] *I've learned new recipes from this"* (survey respondent 11), highlighting the importance of Community-based initiatives with opportunities for sharing knowledge to promote sustainable food access and healthy eating.

3.3.2 Addressing financial barriers to healthy food access

An important barrier identified by the Community that hinders local and traditional food access is affordability. The Community has identified programs that help reduce this barrier to improve accessibility: *"We do freezer workshops. People come in once a month, prepare a meal, and then take them home to cook at no charge"* (key informant 1). Participants voiced that providing healthy foods and learning opportunities at no cost to the Community could improve food access and health outcomes. Other programs, such as free meal replacement beverages to manage blood sugar offered by the Community health center, and a subscription to weekly produce from the farm during the summer, provide opportunities for the Community to have more consistent, affordable access to healthy foods. Additional programs were referred to as a give and take system, highlighting the traditional practice of sharing meals and resources to help one another. A Community member shared an initiative that aims to improve access to healthy and traditional foods for Community members who may need that extra support: *"A Community Fridge Pantry is set up at the Community Hall as a give and take system"* (key informant 4). These types of Community programs are offered to anyone in the Community helping to improve the accessibility of healthy foods by fostering a healthy local environment.

3.3.3 Promoting holistic wellness through land-based practices

Traditional food knowledge and skill development programs were voiced by the Community as a desired resource. Additional programs were identified by Community members that involve engagement with traditional foods and land-based learning: *"Wild Rice planting programs have been initiated along the shores of Chief Island"* (key informant 3). Participants also highlighted that cultural practices are foundations of holistic wellness and are therefore prioritized by the Community. Community members shared that programs such as gardening and harvesting opportunities contribute towards staying culturally grounded, expressing a deep appreciation for these programs that help them stay connected with their cultural roots: *"I really appreciate the seed program Rama has been doing the last few years"* (survey respondent 13). Participants also shared programs such as survival courses for teaching gathering and food preparation skills, as well as guided walks mapping out the medicines within the Community. These land-based education and Community-guided activities help to further reinforce traditional food systems and treaty rights: *"Culture is hosting guided fishing trips to demonstrate to members that fishing and harvesting rights under treaty territory can be upheld."* (key informant 4). While opportunities to engage Elders and knowledge carriers were identified as a key value towards restoring traditional food practices, participants emphasized that some programs are both Community-led and provide opportunities to collaborate with knowledge carriers and Elders within the Community: *"The Health and Culture teams are leading a planting program with direction from an Indigenous Knowledge Carrier"* (key informant 4). These insights highlight the importance of leveraging current programs to uphold intergenerational knowledge sharing and land-based learning to support Community wellbeing.

3.4 Mobilizing existing community assets to strengthen the local food system

This theme encompasses the Community's priority to uplift resources and food champions to support in strengthening the food system. Identifying assets in the Community, can help build towards a sustainable future among all the Williams Treaties First Nations. One of our key informants believes that "*there are many skill sets and passions within our Community*" (key informant 3), particularly knowledge on food production and professional skills (Figure 5). Considering that some barriers to achieving food security are related to a lack of traditional knowledge and food acquisition skills, many individuals who hold this knowledge, such as gardening and fishing skills, can support learning opportunities.

While many diverse programs and initiatives were identified within the Community, barriers to access were shared. Factors such as limited promotion of existing services and spaces, and scheduling concerns that may not accommodate the needs of all Community members. One key informant shared that proper implementation, and evaluation frameworks could help improve access to both existing and future programs: *"It's great to have these ideas, but if you do not have the right steps to implement and evaluate, then they are not successful"* (key informant 1). These insights highlight the need to raise awareness of existing initiatives and consider approaches to assess opportunities for program improvements to achieve greater reach and impact. The strong professional skills in areas such as communication, project leadership, and event planning (Figure 5) reinforce that many individuals within the Community hold valuable strengths that could support the advancement of food initiatives, even without formally holding leadership roles. One key informant further emphasizes this notion when discussing current Community harvesting practices to encourage self-determination: *"If you pick up an apple off the tree on the way home, you are a harvester. If you gather*

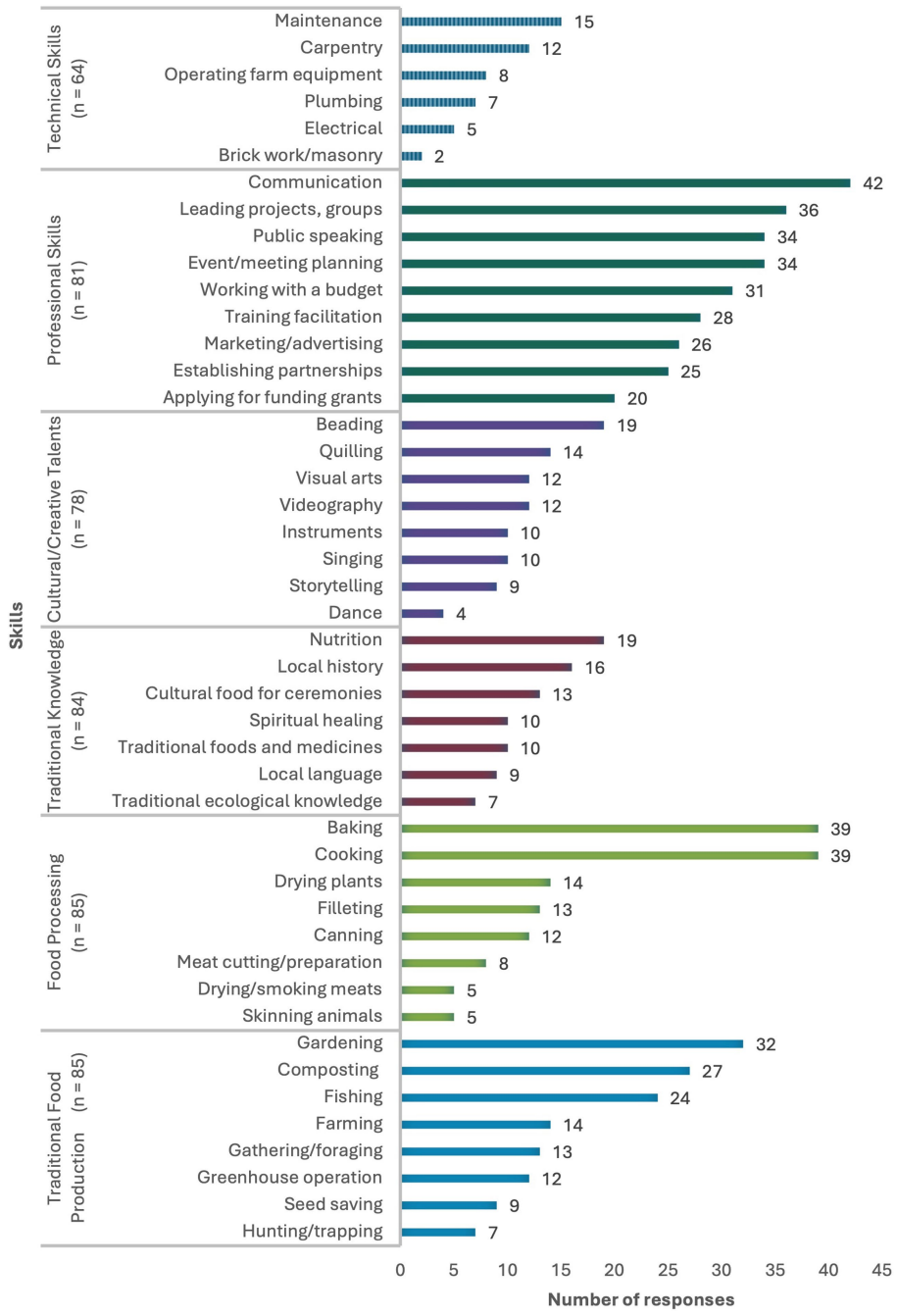


FIGURE 5
Community-identified skills that represent strengths to support food-initiatives.

birch bark, you are a harvester. If you retrieve a porcupine for quills, you are a harvester" (key informant 4). Opportunities to uplift strengths, revitalize land resources, and collaborate with other Communities of the Williams Treaties First Nations were identified as critical to promote Community-led food systems.

3.4.1 Revitalizing land and natural food sources within the community

Participants shared that there are many land-based resources and spaces within the Community that may support current or future programming. Such resources include unused properties and farmland that can support gardening projects or a chicken coop for the Community. By revitalizing these resources, Community members highlighted that this could improve engagement with traditional food practices. The Community feels that some of these resources are being wasted, as they can be used to further support the Community: *"I wish Rama could spend some time cleaning up the apple trees because they are already there and waiting to be put to use*" (survey respondent 9). This highlights the Community's priority for a sustainable food system and using all resources available. A conversation with one of our key informants highlights this long-standing Indigenous principle by encouraging the use of additional unused natural food sources within the Community: *"There are many trees in the Community that can be harvested: choke cherries, pear trees, and apple trees"* (key informant 4).

3.4.2 Fostering shared stewardship with Williams treaties first nations for sustainable food systems

A common consideration within the Community is to explore opportunities for knowledge exchange and collaboration with the Williams Treaties First Nations. Many revealed that shared stewardship will help to strengthen Nation to Nation collaboration and promote stable Indigenous-led food systems. There are opportunities to revive regional structures between Nations such as the Williams Treaties Harvesters Committee for strengthening regional coordination and collaborative frameworks. Community members view expanding beyond Rama First Nation as an opportunity to strengthen the local food system to ultimately access more within and beyond their Community: *"Accessing more in our Community would be amazing through partnering with Nations"* (survey respondent 14). Participants voiced that opportunities for partnerships between the Nations and within the Community will support progress towards the goal of expanding access to resources, programming, and land-based learning. Ideas such as integrating traditional foods within existing food box subscriptions and other programs may expand access and engagement with traditional foods. Rama has been described by the Community as a leader in food sovereignty efforts, such as implementing Community guidelines and policies to encourage more exposure to healthy foods: *"A nutritious food declaration policy was recently implemented to ban unhealthy foods and drinks for community events"* (key informant 1). One of our key informants highlighted the Community's leadership qualities starting with the local farm. They expressed that this asset is an opportunity to share the knowledge gained through the Community's food sovereignty efforts with other Nations to build sustainable food systems: *"The Rama Farm is 100% Community owned and operated. Food is grown by our Community for our Community. A long-term goal is to promote Indigenous grown food. We have a lot of knowledge to share"* (key informant 2).

4 Discussion

Grounded in a Community-led, collaborative, and strengths-based approach, this project highlights existing food-health related programs, and Community challenges and opportunities to enhance the local food system within Rama First Nation. Through Indigenous methodologies of relationality and Two-Eyed Seeing, and western approaches of CBPR, Community's voices and priorities were centered through meaningful engagement between the Community and the project team. Studies have shown that Indigenous food sovereignty efforts are most effective when grounded in relational and locally driven planning frameworks (Herrmann et al., 2020; Desmarais and Wittman, 2014; Prieto et al., 2023). Rather than proposing a top-down strategy, this project weaved Community-identified priorities within implementation strategies to reflect local values, build on existing strengths, and allow for both immediate action and long-term direction. The project was led by a Community member (T.H.) who served as a champion for our Community-guided approach. This facilitated opportunities to engage with Community leadership and the Community Elder for guidance on approaches to knowledge gathering and sharing. Such Community-led engagement served as a crucial component for the strengths-based approach applied to support food systems planning and promote collaboration within Rama First Nation and across the Williams Treaties Territory.

The knowledge gathered identified land-based practices and current initiatives offered in the Community that are associated with a traditional diet. Despite having such resources available in the Community, barriers still impact access, availability, utilization, and stability of the Community's food system. However, there are opportunities to address food insecurity challenges, including a strong desire to reconnect with local and traditional foods with an emphasis on the need for support through education and knowledge sharing, equipment, and financial assistance. Community members highlighted many strengths, skills, and assets within the Community that may help to advance the local food system.

4.1 Food sustainability, food security, and food sovereignty

In response to ongoing food security challenges, Community members articulated a strong commitment to strengthen the local food system and build greater resilience. The Community's perspectives on barriers to accessing local and traditional foods provide direction for meaningful change. These concerns included limited variety of fresh produce and food quality in local stores, high food costs limiting access to local foods, the loss of knowledge on traditional food preparation and skills, and the environmental changes disrupting their food practices. These challenges amplify findings in the literature highlighting a need for responses to address the barriers that have impacted Indigenous food systems and food security (Hanemaayer et al., 2020; Parker et al., 2019; Bruner and Chad, 2014; Wendimu et al., 2018; Skinner et al., 2014). Common challenges to food security faced by Indigenous Communities include high food costs, lack of access to healthy foods, limited retail environment, and limited availability of desired food items in local stores (Parker et al., 2019; Bruner and Chad, 2014; Skinner et al., 2014; Balalio and Robidoux, 2025; Skinner et al., 2013). The loss of traditional knowledge and disrupted connections to the land are commonly reported concerns impeding food security and food sovereignty for Indigenous Communities as well. Factors include the introduction of western foods, restrictive land-management, and disrupted intergenerational knowledge sharing, due to colonial expansion (Kuhnlein et al., 2004; Hanemaayer et al., 2020; Bhawra et al., 2015; Douglas et al., 2014). Such factors impacting food security and food sustainability, also impact food sovereignty.

Underlying the Community's priority to strengthen the local food system is a recognition of the right to food and self-determined food systems. This can be supported through actions that focus on reclaiming land practices, revitalizing skills, and reconnecting with traditional foods and ways of knowing to promote food sovereignty (Coté, 2016; Wittman et al., 2010; Grey and Patel, 2015). The findings from our work have informed several pathways to support harvesting practices, locally produced food, land-based learning, and health-social programming to address nutritional needs and cultural food preferences as part of the Community's food sovereignty endeavours. The lived experiences and perspectives of those from Rama First Nation, extend the literature to underscore the importance of Indigenous food sovereignty described as a principle of "sustainable self-determination" to nurture health by repairing and fostering relationships with the environment (Alfred, 2005; Barker, 2005; Corntassel, 2008). Indigenous scholars Alfred, Barker, and Corntassel further describe that for

Indigenous peoples, "sustainability is intrinsically linked to the transmission of traditional knowledge and cultural practices to future generations. Without the ability of Community members to continuously renew their relationships with the natural world (i.e., gathering medicines, hunting and fishing, basket-making, etc.), Indigenous languages, traditional teachings, family structures, and livelihoods of that Community are all jeopardized" (Corntassel, 2008, p.118). Current strategies within Rama First Nation and newly identified initiatives, reflect the Community's recognition of the importance of food sovereignty in supporting cultural identity, food security, and wellbeing.

While many programs already exist in the Community, several culturally relevant nutrition education initiatives and opportunities for hands-on learning to support skill development in food preparation, harvesting, and building traditional food knowledge were identified. These insights emphasize the priority to promote food knowledge and skills, including the nutritional benefits of a traditional diet. Food-health related programs implemented in Communities, such as nutrition education workshops and activities, have been shown to enable informed decision-making, improve health outcomes, and promote healthy eating behaviors (Ferreira et al., 2022; Gillies et al., 2020; Saksvig et al., 2005; Kolahdooz et al., 2014). In particular, the skill-developing activities in such a program have the potential to support the building of nutrition and traditional food knowledge, which are important for promoting food security and food sovereignty (Shafiee et al., 2022). Engagement with the Community to create an inventory of programs, skills, and food system priorities can support the Community in food system planning, including opportunities to build on food assets to address the challenges to traditional and local food access.

4.2 Deepening community engagement towards community-led strategies

The findings highlighted a broad range of programs, skills, and opportunities that could be mobilized to support food champions and build on existing resources. These findings reflect a shared commitment to drawing on local knowledge, cultural strengths, leadership, and support networks to collaboratively address food security and food sovereignty challenges. Recognizing and creating space for these capacities may further self-determination by enabling Community members to guide and sustain locally driven initiatives. Consistent with other Indigenous food planning efforts, strategies that are flexible, Community-guided, and rooted in relationships are more likely to effectively support actions towards food sovereignty (Prehn, 2025; Ferreira et al., 2022; Delormier et al., 2018; Davies et al., 2023). Overall, the findings point to a strong collective vision for a more intentional and Community-led process that translates these ideas into meaningful and sustained action.

The diverse range of skill sets identified within the Community may be applied to address the gaps within current programs, such as scheduling concerns, and limited promotion of existing services and resources. Developing dedicated roles or staff to help guide planning efforts may improve program accessibility by strengthening outreach, maintaining momentum, and ensuring follow-through. Further engagement with the Community could address these gaps as similar priorities have been noted in other Indigenous-led initiatives, where limited staffing and administrative capacity have created barriers to implementation, despite strong Community support (Balalio and Robidoux, 2025; Ferreira et al., 2022; Peach et al., 2026; Dedyukina et al., 2023). The project findings also revealed underused Community assets with the potential to support current or future land-based programs, expand program capacity, and reduce logistical challenges, including limited space. Revitalizing these resources, could improve engagement with traditional food practices and contribute towards a more sustainable food system (Balalio and Robidoux, 2025; Dedyukina et al., 2023; Sacca et al., 2022). Thus, investing in Community capacity is a practical and necessary approach to support sustainable food system transformation.

The implementation of food-related initiatives is strengthened by consistent and effective communication mechanisms across all levels of community programming. As highlighted in this project, some initiatives desired by Community members, such as learning traditional food knowledge and skills, have already been implemented in different programs. This highlights a potential area for improvement within program development strategies, as well as with communication and awareness. Leveraging current programs to combine priorities such as integrating nutrition education within skill-based programs, or integrating traditional foods within cooking workshops, may address this gap. Moreover, establishing systematic processes for sharing updates, evaluating impact, and identifying opportunities for engagement can reduce duplication, improve accessibility, and increase trust in local planning. As other Indigenous food-planning have shown, embedding clear and reciprocal communication pathways within program implementation and evaluation is critical for collective decision-making and enabling adaptive planning of local food systems (Coté, 2016; Corntassel, 2008; Peach et al., 2026; Domingo et al., 2023). Thus, these outcomes highlight that sustained Community engagement, dedicated resources, and consistent efforts in delivering programs, are necessary to support Community food system priorities.

4.3 Future implications

Advances within this project will involve ongoing engagement to support the Community in food systems planning. This may include providing support in securing resources and in program evaluation activities to strengthen existing initiatives or to identify opportunities for program development. As with other Community-based projects, actions on food security and food sovereignty may require additional planning to integrate Community priorities into existing initiatives such as land use practices, economic development, and Indigenous governance principles to enable action (Ferreira et al., 2022; Gillies et al., 2020; Davies et al., 2023; Peach et al., 2026). Formative work with the Williams Treaties First Nations established common understandings of food security, sustainability, and Indigenous self-determination, providing a strong basis to support collaboration and project coordination across the territory (Domingo et al., 2021a). Further engagement with the Community could provide insight into expanding the Community's vision for improving food and health-related programs.

Next, steps to support the Community will involve raising awareness of existing initiatives, promoting participation, and fostering Nation-to-Nation relationships for collective action on food security and food sovereignty among the Williams Treaties First Nations. Partnering with relevant organizations and aligning Community-identified needs with program development, can help with implementing actions to strengthen the local food system. As access to

healthy food is a key priority for Communities, food is medicine approaches are being explored as an opportunity to integrate traditional foods, practices, and community-based initiatives into local healthcare settings. For example, food prescription programs have been used to promote access to healthy foods and nutrition education through various Community partnerships (Little et al., 2022; Heasley et al., 2021). Gathering Community perspectives on the feasibility of such an approach, could help with informing culturally relevant considerations to meet Community food-health related preferences and interests.

4.4 Strengths and limitations

This project has several strengths and limitations. The timing of our recruitment for key informant interviews presented challenges. Out of respect for the Community, our outreach activities were put on hold when navigating competing priorities and Community circumstances. As a result, only four key informants participated in conversational interviews. Despite these challenges, we recognize that sample sizes in qualitative research are generally small and the selected key informants held diverse expertise about the local food system, nutrition, traditional practices, health, and Community services, providing rich information relevant to our project objectives. Moreover, the open-ended questions in the survey welcomed additional insights from Community members by providing the opportunity to expand on responses. While Community project leads sought to ensure a comprehensive and inclusive recruitment approach for survey participation, no restrictions were placed on the number of responses submitted from the same household. Accordingly, responses were analyzed at the individual level, and findings should not be interpreted as representative of households. While the survey was comprehensive of Community perspectives, it is possible that the length and time to complete the survey may have prevented participation from more Community members. Additionally, most survey questions were intentionally optional, allowing participants to skip items at their discretion. Consequently, sample sizes varied across survey items, which may have influenced interpretation of the findings. However, the survey's flexible design prioritized participant autonomy and supported a community-centered approach by enabling respondents to determine the extent of their participation. This approach likely fostered greater comfort, trust, and supported Community engagement. Moreover, the survey's strengths-based approach accompanied by the open-ended questions and opportunities to provide deeper insight to individual responses, centered Community members voices to capture their values and strengths. This ensured that the findings were rooted in the Community's perspectives aligning well with our project goals. As Indigenous peoples are diverse and Indigenous knowledges are place-based, our findings may not be applicable to other Communities. However, the community-driven and strengths-based approaches undertaken may provide guidance for future research conducted in similar contexts with shared food security and food sovereignty priorities. Despite these limitations, these findings provide valuable insights into the Community's strengths and perspectives on opportunities to improve access to healthy food. Furthermore, the findings provide a foundation on which this broader initiative can grow for food security and food sovereignty efforts across the Williams Treaties First Nations.

5 Conclusion

Through the use of Indigenous methodologies and community-based participatory research, we engaged Rama First Nation Community to inventory existing food-health related programs and identify Community priorities to support food system planning. Current engagement with local and traditional foods presents notable challenges related to access, availability, and utilization, threatening food security and food sovereignty. The findings identify ways to respond to food security through programs and initiatives that support nutrition education, opportunities to learn traditional knowledge and skills, and address financial constraints. There are several opportunities to mobilize food champions and Community assets to further enhance programs to support sustainability. Moreover, deeper engagement could enhance programming by partnering with relevant organizations, aligning Community-identified needs with program development, and enhancing Nation-to-Nation collaboration among the Williams Treaties First Nations. The knowledge gathered highlights the potential of Community governance structures to support food security and food sovereignty through sustainable and culturally relevant practices. The strengths-based approach taken in this initiative can help inform future efforts to integrate Indigenous approaches with Community engagement, which are critical to building shared knowledge for food systems planning. This project demonstrates that a participatory approach to inventory Community strengths can inform relevant responses to food security and food sovereignty that align with Community priorities.

Data availability statement

The datasets presented in this article are not readily available because the Community Partner and Cambium Indigenous Professional Services are the owners and stewards of the data. Requests to access the datasets should be directed to t.copeland@indigenousaware.com.

Ethics statement

This project has been reviewed by the Research Ethics Board at the University of Guelph for compliance with federal guidelines for research involving human participants (REB #1203). The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

TH: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Validation, Writing – original draft, Writing – review & editing. JA: Formal analysis, Writing – original draft, Writing – review & editing, Methodology. MW: Formal analysis, Methodology, Writing – review & editing. K-AC: Conceptualization, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

KS: Methodology, Writing – review & editing, Funding acquisition. AD: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

The remaining author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was not used in the creation of this manuscript.

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Supplementary material

The Supplementary material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fsufs.2026.1876414/full#supplementary-material

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